

SIDHARTHA PULLURI

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INTRODUCTION

Computer Science graduate student at UT Dallas building AI applications, full-stack products, and computer vision systems. Experienced with Python, React, Node.js, PyTorch, TensorFlow, and Hugging Face, with published medicinal plant recognition research and deployed web and Android projects.

EDUCATION

The University of Texas at Dallas

Richardson, TX

Master of Science, Computer Science | GPA: 3.65/4.0

Aug 2025 – May 2027 (Expected)

Relevant Coursework: Natural Language Processing, Artificial Intelligence, Database Design, Advanced Machine Learning

Chaithanya Bharathi Institute of Technology

Hyderabad, India

Bachelor of Science, Computer Science | GPA: 8.7/10.0 (First Class with Distinction)

Aug 2021 – May 2025

Relevant Coursework: Machine Learning, Computer Vision, Deep Learning, Natural Language Processing, Data Structures

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, JavaScript, C++, Kotlin, TypeScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost

NLP & LLMs: Hugging Face Transformers, Sentence-Transformers, LangChain, Gemini API, BERT, GPT, spaCy, NLTK

Computer Vision: OpenCV, YOLO, EfficientNet, EfficientDet, BiFPN, Vision Transformers, Grad-CAM

Tools & Platforms: Git, Docker, Flask, FastAPI, Node.js, Redis, Gradio, Jupyter Notebook, Google Colab, AWS, Weights & Biases, Postman, Vercel

Web & Mobile: React, Vite, Jetpack Compose, Media3 / ExoPlayer, Material 3

Databases: MySQL, PostgreSQL, SQLite, Vector Databases (Pinecone, Chroma)

FEATURED PROJECTS

Medico (CareMatch India) — AI Doctor Recommendation Platform

Mar – May 2026

JavaScript, Node.js, Gemini API, Hugging Face, Google Maps API | [Live website](#)

- Built a medical navigation prototype that parses symptoms and follow-up answers to recommend care level, specialty, and hospital options for Indian cities with INR self-pay cost estimates
- Combined a Gemini medical-language parser with a local multilingual sentence-transformer fallback to handle English, Hinglish, and transliterated symptoms
- Implemented deterministic emergency red-flag checks, Naive Bayes body-system routing, and Google Maps/Places discovery with cost and quality sorting

Returnly — QR Lost & Found Platform

Feb – Apr 2026

Python, Flask, PostgreSQL | [Live website](#)

- Built and deployed Qr-Returnly, a QR-based lost-and-found platform (live at qr-returnly.vercel.app) that turns a single reusable QR tag into a return flow for bags, wallets, badges, keys, and luggage
- Designed an account-based owner dashboard for creating and managing multiple QR tags, with a Lost Mode toggle that updates status without ever regenerating the QR code
- Built a finder scan flow that routes finders to the right contact or reporting step without exposing owner information directly on the physical item

Rule-Based Data Visualization Tool

Python, Flask, Pandas, Matplotlib, Seaborn | [Live website](#)

- Built a deterministic analytics workspace for CSV, TSV, Excel, JSON, and ZIP uploads, supporting nested data, multiple worksheets, and headerless files from parsing through dashboard export
- Implemented schema detection and cleaning for missing values, duplicates, types, and text, with delimiter inference and worksheet diagnostics
- Created an explainable chart advisor with ranked recommendations, a built-in chart guide, and bar, bubble, density, beeswarm, hexbin, and relationship-graph visualizations

YouTube Shorts Story Generator

Python, Flask, Edge TTS | [GitHub](#)

- Engineered a full text-to-video pipeline that converts written stories into short-form vertical videos with AI narration and precisely timed captions
- Automated the entire content pipeline end-to-end, from script ingestion to rendered video, enabling rapid, hands-free content production
- Published a companion browser-based remote panel for the Shorts workflow. [Remote panel](#)

First Move — Hidden Invitations

Python, FastAPI, JavaScript, AES-GCM | [Live website](#)

- Built an invitation-sharing web app that embeds an encrypted message and reveal photo inside a downloadable PNG; recipients upload the image and answer a personal question to unlock it
- Implemented answer-derived encryption, custom cover images, and a responsive reveal experience

TuneVault — Offline Android Music Player

Kotlin, Jetpack Compose, Media3 / ExoPlayer, MediaStore, Material 3 | [Live website](#)

- Built a native Android music player with device and folder-based song discovery, albums, playlists, likes, search, and offline playback without an account
- Integrated MediaStore and the Storage Access Framework for local libraries and published an app download website

Nestly — QR Container Organizer

React, TypeScript, Vite, Supabase, QR Codes | [Live website](#)

- Created a container inventory app with printable QR labels that open a focused view of each container and its contents, connecting physical storage to a searchable digital inventory

Sidhartha Utilities — Everyday Web Tools

React, Vite, Node.js, REST APIs | [Live website](#)

- Built a multi-utility web platform with a responsive task manager, persistent task storage, and create, update, and completion workflows

MACHINE LEARNING PROJECTS

AI-Powered Audio Storytelling System

Aug 2025 – Nov 2025

Python, Bark TTS, NLP

- Built an audio production pipeline for 50+ chapters with automated text segmentation, punctuation normalization, and voice modulation
- Reduced TTS latency by 40% using GPU batch processing while preserving natural prosody across 10+ hour audiobooks
- Assembled chapters with dynamic silence padding and seamless transitions; beta testing with 200+ users achieved 95% satisfaction

Medicinal Leaf Recognition System

Jul 2024 – Mar 2025

PyTorch, EfficientNet, BiFPN, Gradio

- Combined EfficientNetB0 and BiFPN for multi-scale medicinal plant recognition, achieving 94.2% top-1 accuracy on 10,000+ images
- Improved robustness by 12% on mobile captures using geometric transforms, color jittering, and mixup augmentation
- Deployed a Gradio application and REST API with Ayurvedic information, serving 500+ daily predictions with sub-1-second inference

GPT-Powered Conversational AI Chatbot

Feb 2024 – Jun 2024

Hugging Face, Transformers, Flask, Redis

- Fine-tuned GPT-2 (774M parameters) on 50,000+ domain dialogues, reducing perplexity by 28% and improving contextual relevance
- Implemented JWT authentication and Redis conversation state with context retention across 10+ exchanges
- Deployed a Flask REST API with load balancing and monitoring, achieving 99.5% uptime while handling 1,000+ concurrent users

PUBLICATIONS

BiFPNet: Bidirectional Feature Pyramid Network for Medicinal Leaf Identification with Therapeutic Usage

International Journal of Computers and Applications | Published Dec 2025 | DOI: 10.1080/1206212X.2025.2608641

Coauthored journal paper combining BiFPNet-based computer vision, Grad-CAM explainability, and therapeutic knowledge retrieval across 98 plant species

LEADERSHIP & EXPERIENCE

Placement Coordinator, Chaithanya Bharathi Institute of Technology

Jun 2024 – May 2025

- Managed end-to-end technical recruitment for 200+ Computer Science students, coordinating placement drives with 30+ technology companies
- Acted as liaison between students, faculty, and recruiting teams to streamline interview scheduling, shortlisting, and communication workflows

CERTIFICATIONS

- Machine Learning and Image Processing – L&T Technologies & NASSCOM Foundation, Mar 2024
- Machine Learning Specialization – Internshala, Jun 2023

ACHIEVEMENTS

- CBIT Hackathon 2024 – 3rd Place: Led a cross-functional team of 4 to build an ML-powered health diagnostics tool using Random Forest and XGBoost, competing against 50+ teams in a 48-hour sprint
- Member: AI BUZ Club (UTD), Artificial Intelligence Society (UTD), Neural Nexus Club (CBIT)